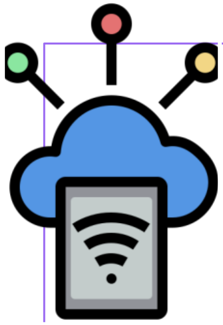
IoT Data Lifecycle

IoT Data Lifecycle

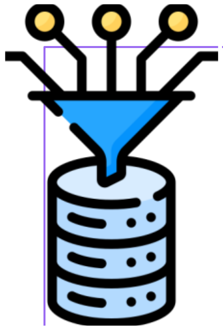
- IoT is a data-driven system
- Every IoT solution follows a data lifecycle
- Architecture decisions depend on data flow
- Lifecycle stages drive cost, performance & security

IoT Data Lifecycle



Data Generation (Sensors & Devices)

Sensors generate raw data
Data can be periodic, event-driven, or continuous
Examples: temperature, vibration, GPS, video
Data quality starts at the device level



Data Ingestion & Connectivity

Data moves via networks
Protocols: MQTT, HTTP, CoAP
Connectivity types: Wi-Fi, Cellular, LPWAN
Reliability & bandwidth matter

Data processing

Edge vs Cloud Processing

Edge processing = closer to device
Cloud processing = centralized intelligence
Latency, cost, and bandwidth trade-offs
Hybrid architectures are common

Data Storage & Retention

Not all data needs long-term storage
Hot, warm, and cold data
Compliance & audit requirements
Cost vs value of data
After processing, data may need

Analytics, AI & Decision Making

Real-time vs batch analytics
AI models learn from historical data
Predictive & prescriptive insights
Business value is created here

Action, Automation & Feedback Loop

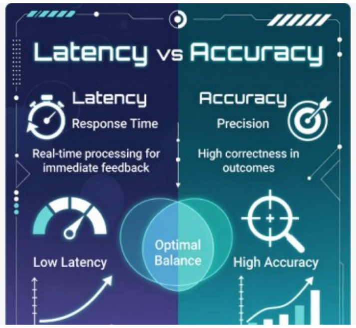
Actions via actuators
Alerts, automation, or human intervention
Closed-loop systems
Continuous improvement cycle

Architectural Trade-offs – Summary

Every IoT architecture is ultimately a balancing act.

Edge vs Cloud

Cost vs performance



Latency vs accuracy

Security vs accessibility

Simplicity vs scalability

There is no single "correct" IoT architecture. The right design is the one that fits the use case, the environment, and the business goals. By understanding the IoT data lifecycle and the trade-offs at each stage, engineers can make informed decisions that align technology with real-world needs.